Ibadan Land-Cover Change and Urban Expansion, 2013-2023

A planning-focused geospatial analysis of metropolitan land-cover change

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Project overview

This project examines how the Ibadan metropolitan landscape changed between 2013 and 2023, with particular attention to the scale of urban expansion and the land-cover types converted as the built-up footprint grew. The analysis covers 3,220.581 km² and maps four broad classes: Built-up, Vegetation, Water and Bare soil.

Planning question

How did Ibadan's land cover change between 2013 and 2023, and what types of land were converted as the city expanded?

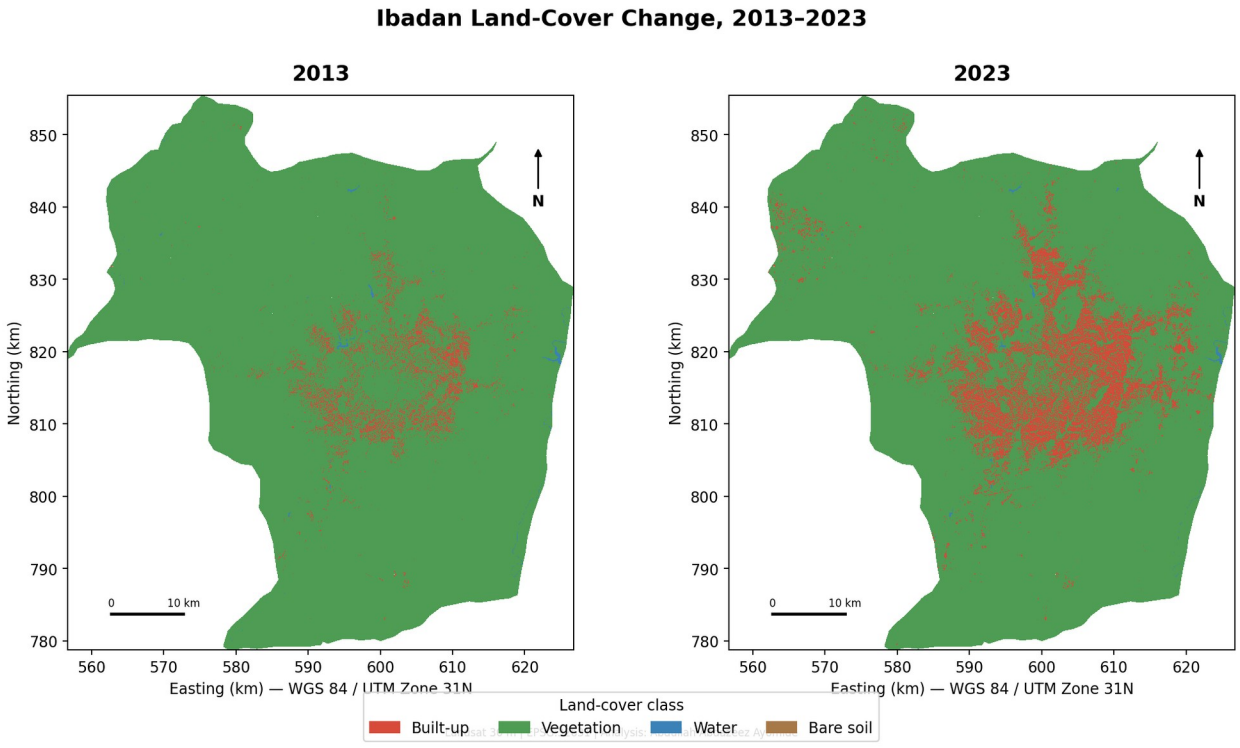


Figure 1. Land-cover comparison for 2013 and 2023.

Data and method

Landsat Collection 2 Surface Reflectance imagery provided six spectral bands (SR_B2-SR_B7), combined with NDVI, NDBI and MNDWI. Both years were aligned to the same 30 m grid in WGS 84 / UTM Zone 31N (EPSG:32631). A Random Forest classifier mapped the four classes, followed by pixel-by-pixel post-classification change analysis.

The final reviewed reference set contained 48 samples: 32 calibration samples and 16 independent holdout samples. The holdout was kept separate from fitting and model-selection decisions.

Validation

Indicator	Result
Correct holdout predictions	14 of 16
Overall Accuracy	0.8750
Balanced Accuracy	0.9259
Macro F1	0.6354
Cohen's Kappa	0.7935

Because the holdout is small, the raw 14/16 result should be interpreted alongside the percentage-based metrics.

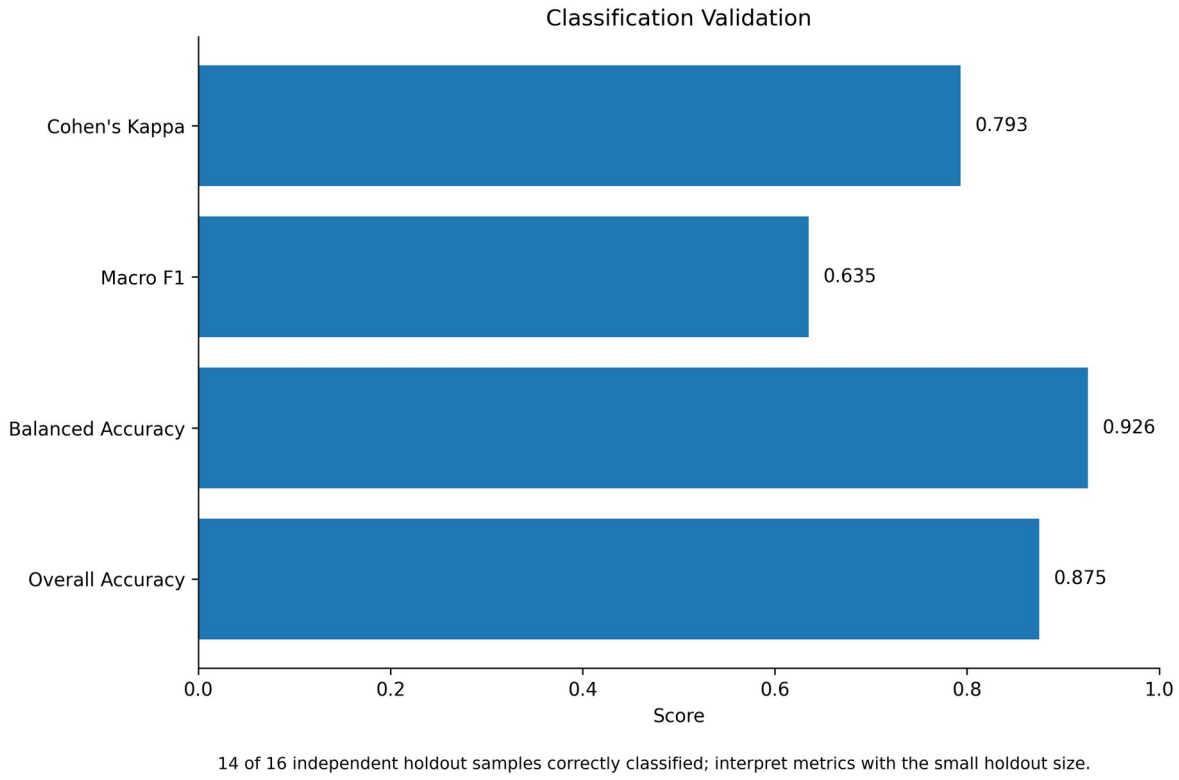


Figure 2. Independent holdout validation summary.

Results

Indicator	Result
Analysis area	3,220.581 km²
Built-up, 2013	99.866 km² (3.101%)
Built-up, 2023	330.177 km² (10.252%)
Net built-up increase	230.311 km²
Relative built-up increase	230.62%
Vegetation → Built-up	246.104 km²
Gross new built-up	248.235 km²
New built-up from vegetation	99.14%
Stable landscape	91.50%
Changed landscape	8.50%

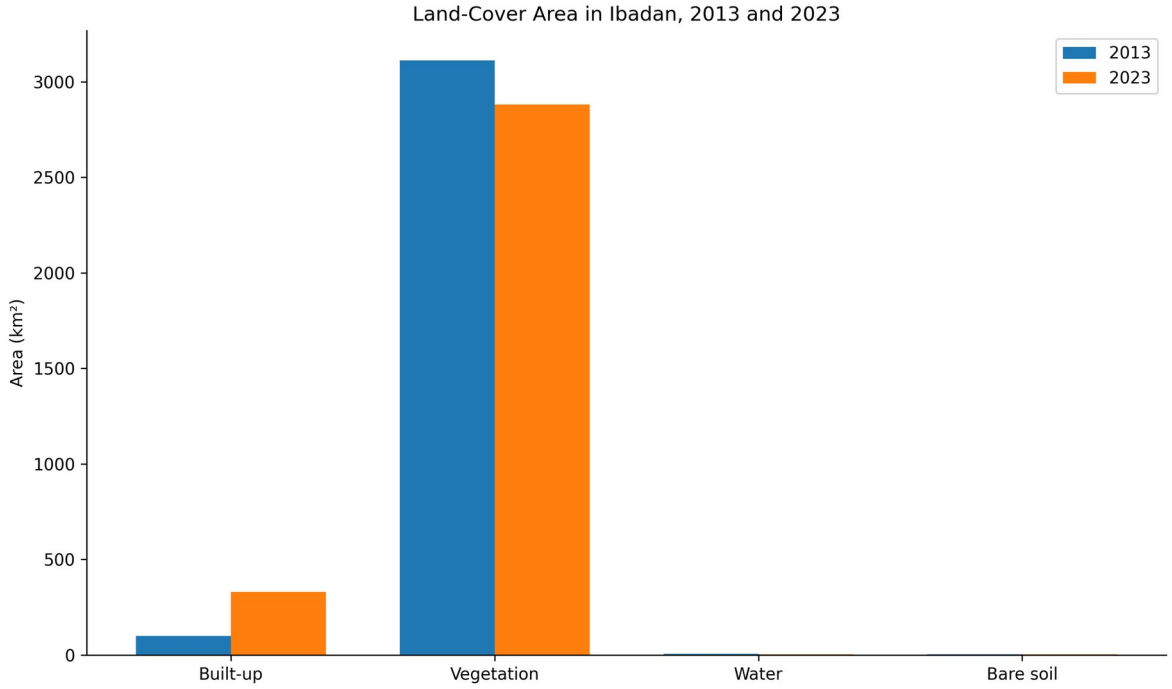


Figure 3. Land-cover area comparison.

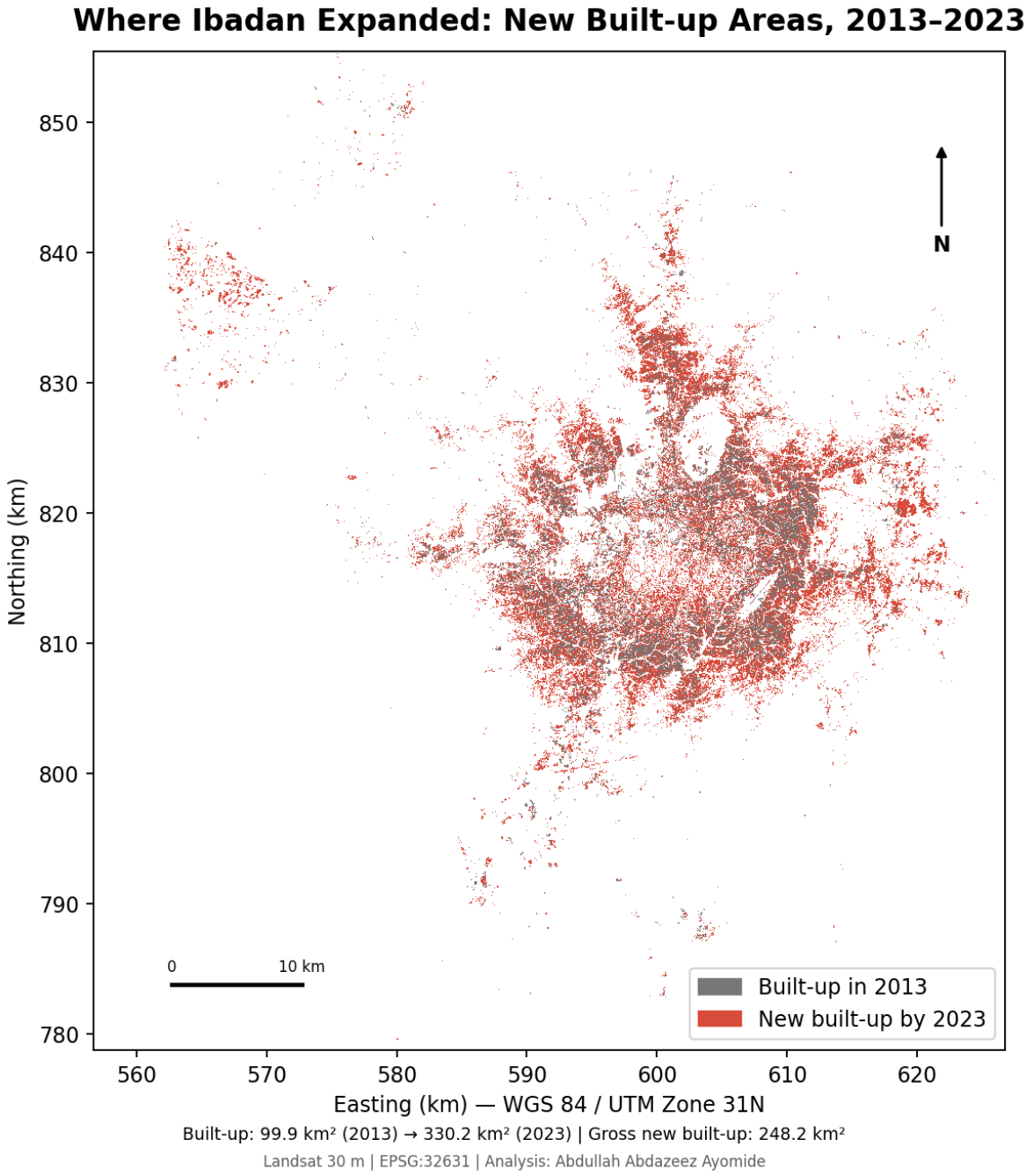


Figure 4. Spatial pattern of built-up expansion.

Built-up land increased from 99.866 km² in 2013 to 330.177 km² in 2023. Vegetation declined by 230.331 km², and the dominant mapped transition was Vegetation to Built-up at 246.104 km². Of 248.235 km² of gross new built-up land, 99.14% had been classified as vegetation in 2013.

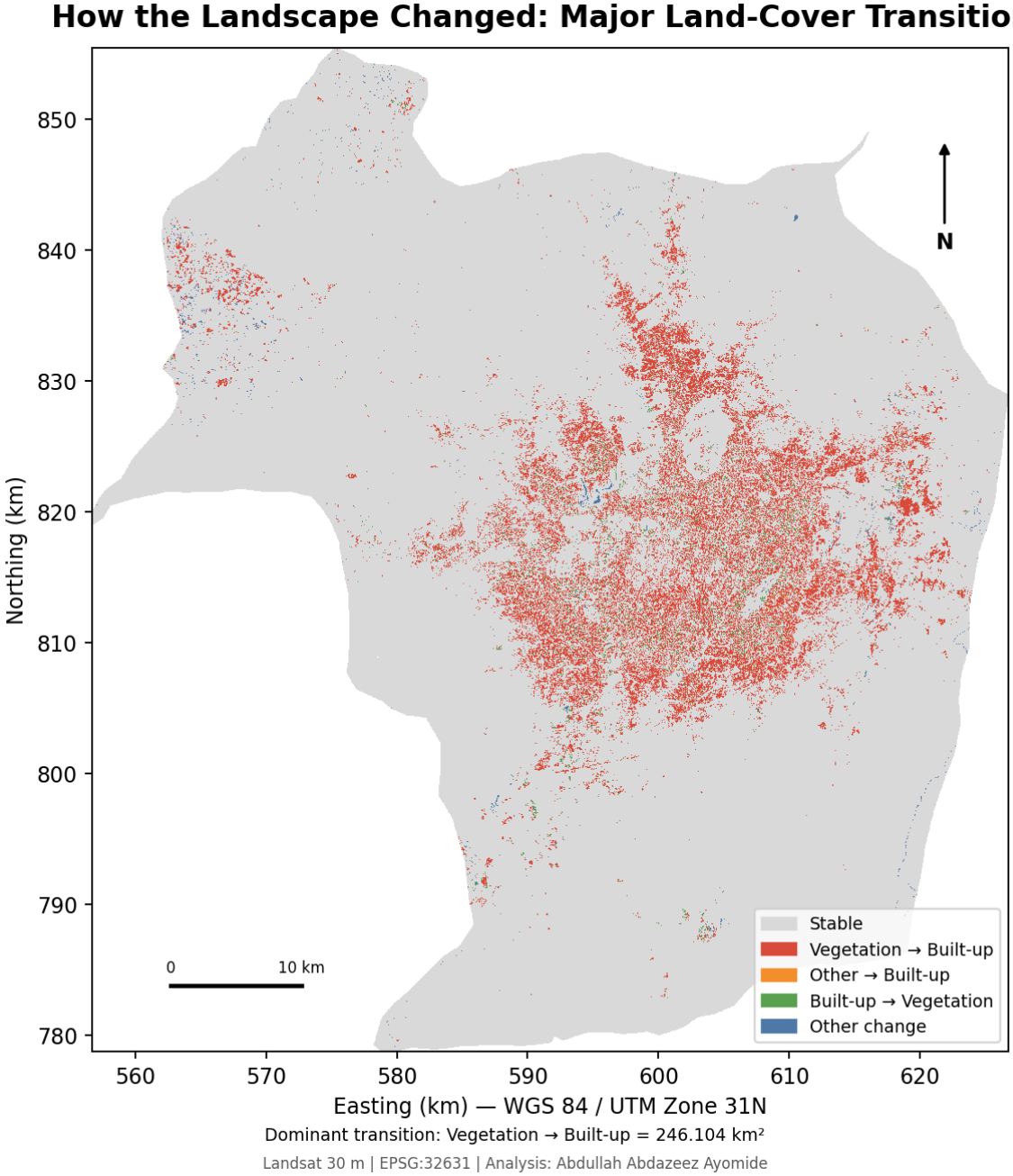


Figure 5. Major land-cover transitions.

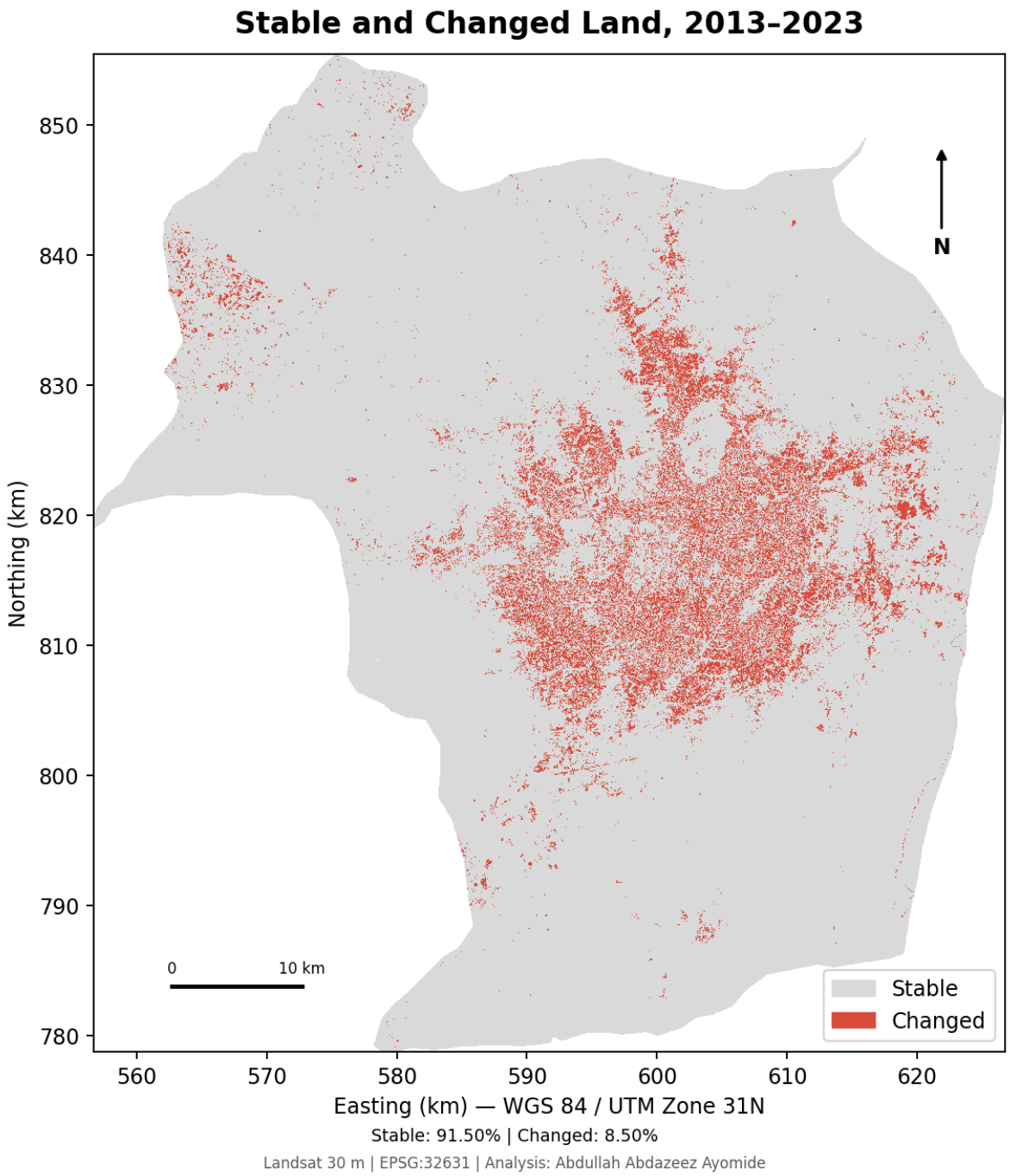


Figure 6. Stable and changed land between 2013 and 2023.

Planning interpretation

Ibadan's expansion between 2013 and 2023 was characterised primarily by conversion of vegetated land at the urban edge into built-up development. The pattern highlights where future growth management, infrastructure coordination and green-space protection will become increasingly important.

- Monitor urban growth fronts regularly.
- Coordinate infrastructure provision with new development.

- Integrate green-space protection into metropolitan expansion strategies.
- Repeat remote-sensing updates periodically to support development management.

Limitations

- The independent holdout contains only 16 samples.
- Landsat's 30 m pixels may mix buildings, roads, vegetation and exposed soil.
- The four-class system simplifies a more detailed metropolitan landscape.
- Spectral consistency does not replace independent ground-reference evidence.
- Land-cover transitions do not establish demographic, economic or regulatory causes.

The results are therefore best used for metropolitan-scale land-cover change and urban-growth interpretation, not parcel-level development decisions.

Tools

Google Earth Engine · Landsat Collection 2 Surface Reflectance · Python · Rasterio · GeoPandas · NumPy · scikit-learn · Matplotlib · GIS · Git · GitHub

Citation

Abdullah Abdazeez Ayomide (2026). Ibadan Land-Cover Change and Urban Expansion, 2013-2023.